Cooperative Underwater Glider Operations supported by a Companion Autonomous Surface Vehicle

An Application of Reinforcement Learning Techniques

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The Markov Decision Process

State Space: A 20-dimensional vector normalized element-wise to $[-1, 1]$ that includes information about the ASV state, the estimated relative states of both gliders, and a shared glider-heading term.

Idx.	Category	Quantity	Normalization
0–2	ASV state	heading ψ, previous surge a_u, previous yaw rate a_r	ψ/π, commands in $[-1, 1]$
3–9	Glider 1	distance $\hat{d}_1$, range rate $\dot{\hat{d}}_1$, relative position $\Delta x_{\mathrm{I}1}, \Delta y_{\mathrm{I}1}$, uncertainty ellipse $(\sigma_{1,1}, \sigma_{1,2}, \theta_1)$	scaled by L, 3.0, L, and σ_s
10–16	Glider 2	same quantities for glider 2	same scaling as glider 1
17	Shared glider info	estimated glider heading $\hat{\psi}_g$	$\hat{\psi}_g/\pi$
18–19	Local current	ocean current at ASV position (c_x, c_y)	scaled by $\mathbf{current}_{\max}$, clipped to $[-1, 1]$

Action Space: A two-dimensional continuous vector $a = [a_u,\ a_r]^\top \in [-1,\ 1]^2$, representing the normalized commanded surge and yaw rate of the ASV. These are mapped to physical commands via

$$u = a_u \cdot u_{\max}, \tag{1}$$
$$r = a_r \cdot r_{\max}. \tag{2}$$

Transition Dynamics: At each time step, the environment updates the ASV, propagates both gliders, and updates the EKF belief state.

- **ASV motion:** the ASV follows a planar unicycle model driven by commanded surge u and yaw rate r:

$$\psi_{k+1} = \mathrm{wrap}(\psi_k + r_k \Delta t), \quad x_{k+1} = x_k + u_k \cos(\psi_{k+1}) \Delta t, \quad y_{k+1} = y_k + u_k \sin(\psi_{k+1}) \Delta t.$$

- **Glider motion:** each glider follows straight-line, constant-velocity motion:

$$p_{i,k+1} = p_{i,k} + v_i \Delta t.$$

- **EKF belief update:** For each glider i, the state and covariance are predicted at time step k and corrected using a range measurement when $d_i \le R_{\mathrm{comm}}$:

$$\hat{x}^-_{i,k} = F \hat{x}_{i,k-1} + B c_{i,k-1}, \qquad P^-_{i,k} = F P_{i,k-1} F^\top + Q,$$
$$\hat{x}_{i,k} = \hat{x}^-_{i,k} + K_{i,k} \left[z_{i,k} - h\left(\hat{x}^-_{i,k}\right) \right], \quad P_{i,k} = (I - K_{i,k} H_{i,k}) P^-_{i,k} (I - K_{i,k} H_{i,k})^\top + K_{i,k} R K^\top_{i,k}.$$

Here, $\hat{x}_{i,k}$ and $P_{i,k}$ are the estimated state and covariance, F is the state-transition matrix, Q is the process-noise covariance, $z_{i,k}$ is the measured range, $h(\cdot)$ is the range model, $H_{i,k}$ is its Jacobian, $K_{i,k}$ is the Kalman gain, and R is the measurement-noise covariance.

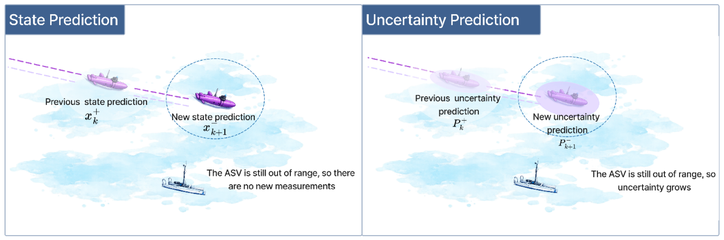


Figure 1. EKF state and uncertainty prediction, occurs at every timestep of the simulation. This prediction is updated when the ASV enters the comm range of the glider.

Reward Function: The reward encourages the ASV to reduce the **D-optimality** criterion, given by the determinant of each glider's position covariance matrix, for both gliders while remaining within acoustic range and avoiding aggressive control. By taking the minimum uncertainty reward across the two gliders, the objective is dominated by the glider with the poorer estimate, encouraging the ASV to switch its attention between gliders as their relative uncertainties change.

$$r_k = w_{\mathrm{unc}} r_{\mathrm{unc}} + w_{\mathrm{comm}} r_{\mathrm{comm}} + w_{\mathrm{ctrl}} r_{\mathrm{ctrl}},$$

where

$$r_{\mathrm{unc}} = \min_{i \in \{1,2\}} \left[-\ln \det \left(P^{(i)}_{\mathrm{pos}} \right) \right], \qquad r_{\mathrm{comm}} = -\sum_{i=1}^{2} \max\left(0, \frac{\|p_{\mathrm{ASV}} - p_i\| - R_{\mathrm{comm}}}{R_{\mathrm{comm}}}\right), \qquad r_{\mathrm{ctrl}} = -(a_u^2 + a_r^2).$$

Reinforcement Learning Architecture

Soft Actor-Critic (SAC) was used as the learning algorithm because it is designed for continuous-control tasks and supports stable off-policy training. Its stochastic actor promotes exploration, while twin critic networks reduce overestimation bias during value estimation. Entropy regularization helps the policy remain robust instead of converging too early to a single deterministic tracking behavior.

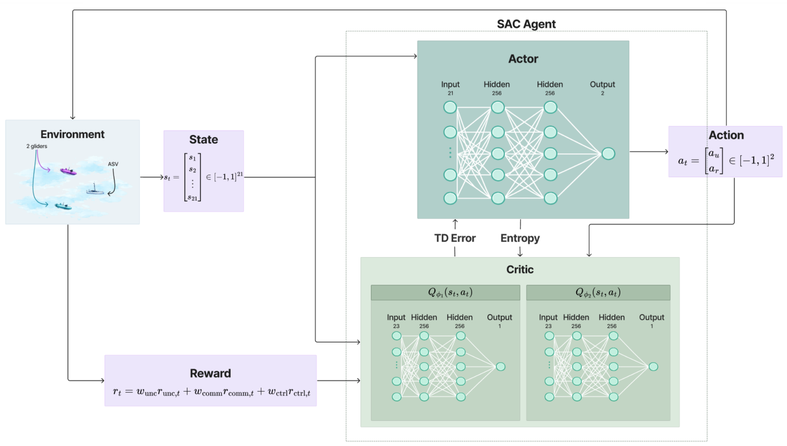


Figure 2. SAC training architecture.

Other RL algorithms tested did not converge during training: Deep Deterministic Policy Gradient (DDPG), Twin Delayed Deep Deterministic Policy Gradient (TD3), and Proximal Policy Optimization (PPO).

Motivation

Underwater gliders enable long-duration ocean monitoring, but their navigation accuracy is limited by the lack of GPS underwater and the cost of high-grade onboard navigation systems. A companion Autonomous Surface Vehicle (ASV) equipped with GPS and USBL can act as a mobile reference, periodically localizing submerged gliders through **range-only** measurements. In multi-glider missions, the ASV cannot always remain within acoustic range of every vehicle simultaneously. This turns ASV guidance into a **single-agent, multi-target reinforcement learning problem** and motivates the central research question:

> How should an ASV move in order to maintain accurate position estimates for multiple underwater gliders when it cannot observe all gliders at once?
>
> The ASV must balance: [1] reducing localization uncertainty, [2] staying within acoustic communication range, [3] limiting control effort.

Resulting Trajectory

The policy was evaluated in a custom OpenAI Gym 2D horizontal simulation environment with two parallel underwater gliders and a companion ASV. The ASV has a maximum surge speed $2.5\ \mathrm{m/s}$ and yaw rate $0.5\ \mathrm{rad/s}$, while each glider moves at $0.30\ \mathrm{m/s}$. Acoustic updates are available within an $800\ \mathrm{m}$ communication radius, and training included spatially varying currents up to $0.20\ \mathrm{m/s}$.

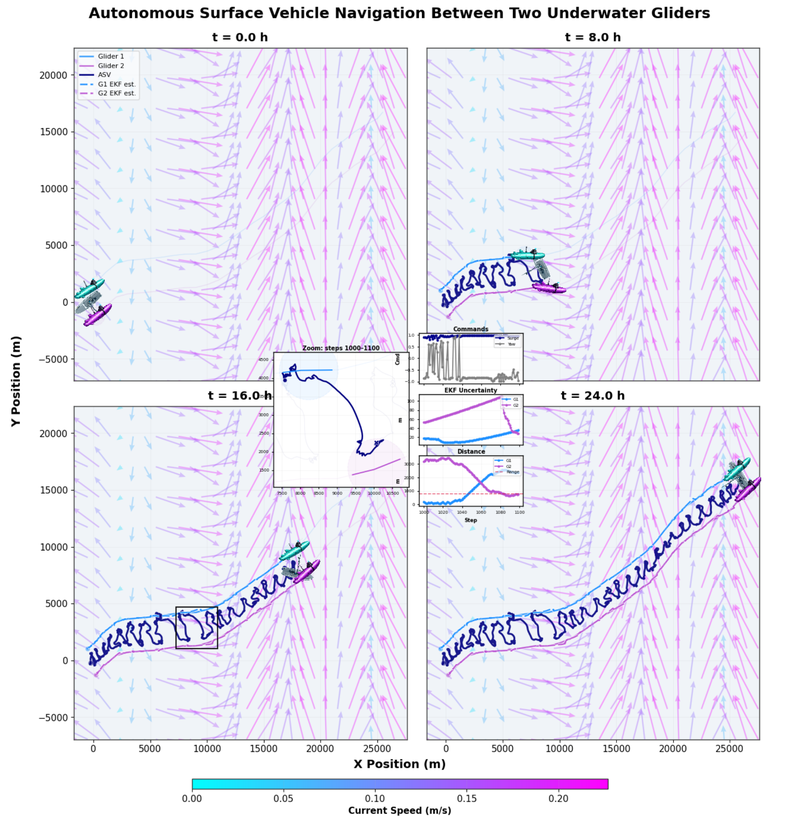


Figure 3. Learned ASV trajectory showing the vehicle alternating between two underwater gliders to maintain acoustic contact and reduce localization uncertainty.

- **Sweeping-arc tracking:** inside a glider's acoustic range the ASV moved in arcs, creating the changing relative geometry that range-only measurements need to keep each glider's EKF uncertainty low (Fig. 3).
- **Alternating, asymmetric service:** the ASV orbits between the two gliders, handing off once one is well-localized and the other's uncertainty has grown (crossing distance/uncertainty traces, zoom inset); coverage is not symmetric, reflecting the randomized initial estimates and per-glider EKF noise (Fig. 3).
- **Bounded error, rising reward:** EKF uncertainty and position error recover after each excursion instead of diverging, while the total D-optimality reward ($-\ln \det P$, the log-determinant of the Fisher information) increases and then stabilizes over the 24 h mission (Fig. 4).

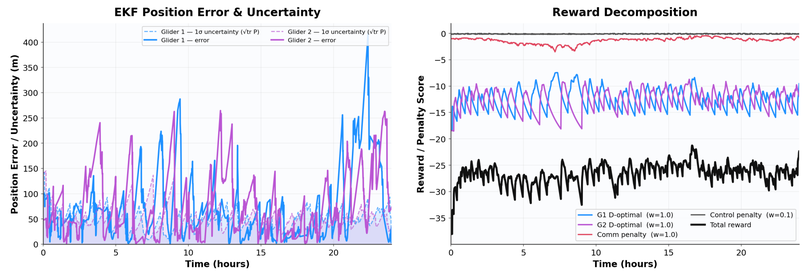


Figure 4. EKF position error and the filter's 1σ ($\sqrt{\mathrm{tr}\, P}$) uncertainty over time for each glider (left), and the per-step reward broken into its weighted components: per-glider D-optimality, communication penalty, control penalty, and the total (right).

Traditional Method Comparison

The SAC policy was compared against a deterministic ping-pong baseline that alternates between gliders using EKF-estimated positions. Across the 24-hour mission, SAC achieved substantially lower localization uncertainty and position error while maintaining acoustic contact with both gliders more consistently. The ping-pong policy's direct, straight-in approach provided too little geometric diversity for its range-only measurements to be informative, causing the tracking to fail.

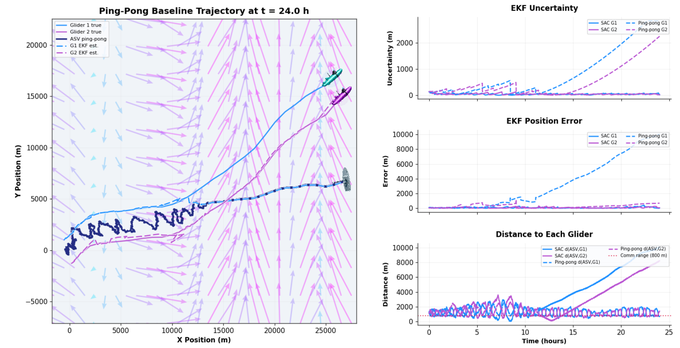


Figure 5. Comparison of the learned SAC policy and deterministic ping-pong baseline. The left panel shows the final 24-hour ping-pong trajectory under spatially varying currents, where the ASV follows EKF-estimated glider positions but is advected away from both true glider tracks. The right panels compare EKF uncertainty, EKF position error, and ASV distance to each glider for SAC and ping-pong. SAC maintains substantially lower uncertainty and position error while keeping more consistent acoustic contact with both gliders.

Metric	SAC G1	SAC G2	Ping-Pong G1	Ping-Pong G2
Mean EKF uncertainty (m)	50.8	53.5	831.6	510.0
Mean EKF position error (m)	75.2	77.7	3250.1	287.9
Time in comm. range (%)	24.5	28.1	1.6	10.2

Compared with the ping-pong baseline, SAC reduced the average EKF uncertainty across both gliders by **92.2%** and reduced the mean EKF position error by **95.7%**. SAC also increased the average time spent within acoustic communication range from **5.9%** to **26.3%**.

Robustness to Ocean Currents

To evaluate current adaptability, the trained SAC policy was tested under different maximum ocean current speeds while keeping the training condition at $0.20\ \mathrm{m/s}$. Each current speed was evaluated over $N = 5$ independent random-seed rollouts; reported values are the seed means and the shaded bands in Fig. 6 denote ± 1 standard deviation across seeds. The policy performed best near the training condition, achieving a mean EKF uncertainty of $45.3\ \mathrm{m}$ and maintaining acoustic communication range for 50.0% of the mission.

Condition	Current	Uncertainty	Comm. range
No currents	$0.00\ \mathrm{m/s}$	$56.8\ \mathrm{m}$	21.2%
Training condition	$0.20\ \mathrm{m/s}$	$45.3\ \mathrm{m}$	50.0%
Max tested current	$0.60\ \mathrm{m/s}$	$918.5\ \mathrm{m}$	25.5%

The policy remained functional up to $0.35\ \mathrm{m/s}$, corresponding to $1.75\times$ the training current speed. However, localization performance then degraded under stronger currents, with sharp uncertainty increases occurring. Communication coverage first dropped more than 10 percentage points below the training condition at $0.35\ \mathrm{m/s}$.

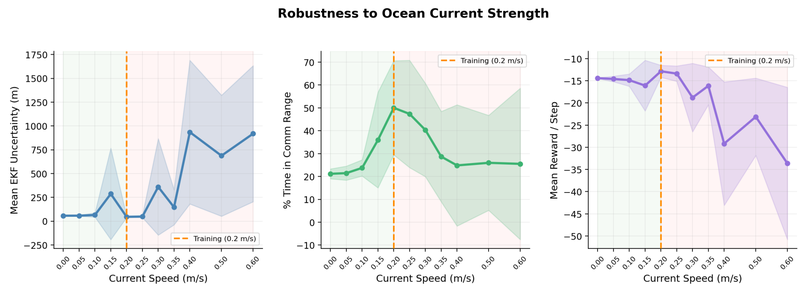


Figure 6. Robustness of the trained SAC policy to increasing ocean current strength. The dashed vertical line marks the training current speed of $0.20\ \mathrm{m/s}$. Performance is strongest near the training condition, while higher current speeds increase EKF uncertainty and reduce reward. The policy remains operational up to $0.35\ \mathrm{m/s}$, demonstrating partial generalization beyond the training current.

Conclusions & Future Work

- **SAC learned an adaptive ASV guidance strategy** that moves between two underwater gliders to reduce EKF localization uncertainty while balancing acoustic contact and control effort.
- **The learned policy outperformed the ping-pong baseline,** reducing average EKF uncertainty by **92.2%** and mean EKF position error by **95.7%** across both gliders.
- **The ASV maintained more balanced multi-glider tracking,** increasing average time within acoustic communication range from **5.9%** for ping-pong to **26.3%** for SAC.
- **Robustness testing showed partial generalization to stronger currents,** with the policy remaining operational up to $0.35\ \mathrm{m/s}$, or $1.75\times$ the training current speed.
- **Future work:** Extend the framework to 3D vehicle dynamics, larger glider fleets, and field-realistic ocean current forecasts.

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